

An Autonomously Navigation System for Forestry Quadrotor within GPS-denied Below-canopy Environment*

Yu Xia, Ming Song, JunGuo Zhang, ChunHe Hu

Abstract—This paper presents a navigation system that enables quadrotor to navigate autonomously within GPS-denied below-canopy environment without GPS. The navigation system consists of the function of Simultaneous Localization and Mapping(SLAM) and path planning. The online real-time locating estimation is achieved by Semi-direct Visual Odometer(SVO) to calculate the odometer data; we locate the position using the fusion of laser radar data and the visual odometer in the front end, the graph-based optimization is used as an optimization mechanism to map in the back end. The path planning finds an obstacle-free trajectory based on the waypoints generated by the improved RRT* algorithm fitted with the Bezier curve.

Keywords—Quadrotor; Below-canopy environment; SVO; Graph Optimization; Improved RRT*

I. INTRODUCTION

The application of quadrotor in forestry has been more extensive for its simple operation[1-2]. Lack of GPS signal and complex illumination interferences are big problems which have brought difficulties to navigate and control quadrotor autonomously within GPS-denied below-canopy environment compare to navigating above the canopy. It is the key issues for quadrotor to navigate within below-canopy environment so that we can build the map and plan the path [3].

The classic SLAM algorithms face more challenges used in complex GPS-denied below-canopy environment[4-5]. Traditional V-SLAM based key-frame is far less robust in changed scene; Hector SLAM included in the Laser-SLAM along with Gmapping requires laser sensor with high frequency and low measurement noise which is usually expensive and heavy. The unstructured environment of the forest area limits the navigation ability of quadrotor[6]. The problems of high density leaf occlusion and illumination difference have great influence on UAV trajectory planning and obstacle avoidance.

This paper proposed a SLAM algorithm based on fusion of monocular vision odometer and laser radar to achieve quadrotor's position and map information. On the basis of

map information, this paper used an advanced RRT* algorithm[7-8] to generate a series of waypoints used to create a smooth and obstacle-free trajectory fitted with Bezier curve.

II. SLAM BASED ON FUSION OF MONOCULAR VISION AND LASER

A SLAM algorithm consisted of the front-end and the back-end[9] has been proposed based on the fusion of monocular vision odometer and laser for sake of navigating the quadrotor conveniently. We use the ICP[10] to fuse the data of laser and vision odometer and build the map by searching the table and Gauss transformation at the front end to increase the locating accuracy; the Graph optimization is used to optimize the map to improve the optimization efficiency at the back end.

This chapter introduces the proposed SLAM algorithm in detail. The location algorithm based on SVO is proposed to solve the problem of weak GPS signal and the main steps are: 1. Estimation of initial position; 2. Optimization of feature point position; 3. Optimization of world posture. The fusion algorithm provides the obstacle map for quadrotor flying in the GPS-denied below-canopy environment and the main steps are: 1. Scan matching; 2. Loop closure; 3. Global optimization.

A. Location Algorithm based on SVO

Visual odometer is usually divided into direct method[10] and feature method. The algorithm of monocular vision based on semi-direct method matches by extracting sparse feature points[11].

This method is called semi-direct method for combining the direct method and the feature method. The flow chart is shown in Fig.1

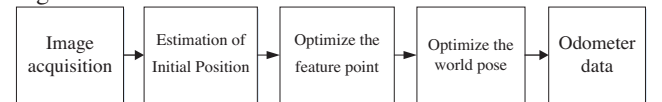


Fig. 1: The flow chart of the Location Algorithm based on SVO

1) Estimation of Initial Pose

Most of the visual odometer methods use sparse feature points to represent camera position and pose, the steps are as follow.

1. Initialization. Supposing the relative position $\hat{T}_{k,k-1}$ between two adjacent frames is known, the feature point's pose (u,v) and depth d can be calculated in the frame I_{k-1} by multi frame feature detection and depth estimation

2. Re projection. Assuming that a common feature's pose (u,v) and the depth d are known. We transform the current frame I_{k-1} to get the three-dimensional coordinate p_k .

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Then the (u, v) is transformed to (u', v') in the I_k through camera internal parameter.

3. Iterative optimization, the (u', v') got by re projection is inaccurate and the brightness values are not equal before and after the projection because the $\hat{T}_{k,k-1}$ is an assumed value in the stage of initialization. We build an optimal function to get the optimized pose $\hat{T}_{k,k-1}$.

$$T_{k,k-1} = \underset{T}{\operatorname{argmin}} \frac{1}{2} \sum_{i \in R} \left\| \delta I(T_{k,k-1}, u_i) \right\|^2 \quad (1)$$

$\delta I(T, u) = I_k(\pi(T \pi^{-1}(u, d_u))) - I_{k-1}(u)$ represents the inter frame optimization functions, $\pi()$ is the projection model function of the camera, $\pi^{-1}()$ is the inverse projection function, the iterative increments of pose can be calculated by the equation (2).

$$J^T J \xi = -J^T \delta I(0) \quad (2)$$

2) Optimization of Feature Point Pose

Since the first step to estimate the initial camera pose is not the most accurate, so the feature point of the projection pose which exists residuals is not coincided with the real feature points. So The theory of illumination invariant is used to construct the objective function of residual invariant feature point position optimization. This method that is similar to the previous step using Gauss Newton method to solve the optimization objective function optimizes the location of the feature points to further constrain the pose of the current frame to avoid the cumulative error caused by estimation of the predicted pose. The optimization function is shown as follow.

$$u'_i = \underset{u'_i}{\operatorname{argmin}} \frac{1}{2} \| I_k(u'_i) - A_i I_r(u_i) \|^2 \quad (3)$$

The equation is the nonlinear least squares problem which has the same iterative calculation equation with the equation(1), the Jacobi matrix is changed as equation (4)

$$J = \begin{bmatrix} \frac{\partial r}{\partial u'} & \frac{\partial r}{\partial v'} \end{bmatrix} \quad (4)$$

3) Optimization of World Posture

More accurate feature points can be got in I_k and we will optimize the camera pose along with the features points' coordinates in the world coordinate system(WCS).

We can optimize the camera's pose $T_{k,w}$ in the WCS by optimize the error generated by re projection of the pixel. The optimization function is shown as follow.

$$T_{k,w} = \underset{T_{k,w}}{\operatorname{argmin}} \frac{1}{2} \| I_k(u_i) - I_k(\pi(T_{k,w}, p_i)) \|^2 \quad (5)$$

The calculated pose can be as the input of the quadrotor's position and pose controller, also be used as odometer information for building the obstacle map by the laser.

B. Mapping for Laser Fusion Monocular Odometer

The V-SLAM algorithm is easily disturbed because of the complex illumination condition within GPS-denied below-canopy environment[12]. The Laser SLAM also has it

own limitation in the smaller scene. According to the above problems, it is necessary to design a fusion of Laser SLAM and V-SLAM to increase the navigation system's robustness. The flow chart of the proposed mapping algorithm is shown as Fig.2.

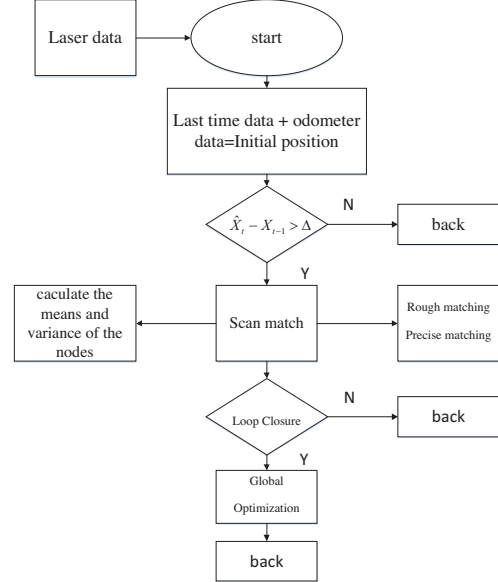


Fig.2: The flow chart of the proposed mapping algorithm

4) Scan Matching

Supposing the quadrotor gets two frame data z_{i-1} , z_i corresponding to the two different poses x_{i-1} , x_i where the quadrotor scans the things. We defines some parameters: u is used to describe the change of the motion, m is the observation value of the observation point in the WCS, z is the value of the distance between the laser and the observation point[13].

If every laser echo z_j is independent, the observation equation is shown as equation (6).

$$p(z | x_i, m) = \prod_j p(z_j | x_i, m) \quad (6)$$

We can get the current estimated attitude angle along with a search table including n angles, and map the laser information represented by robot coordinates in different angles with a certain resolution and offset value according to the odometer. The amount of n can be expressed by the equation (7).

$$n = \frac{angle_{bias}}{angle_{resolution}} + 1 \quad (7)$$

Based on the grid map of the robot coordinate system, the discrete search $n \times n$ area is used to locate the map. The search table matching process is as follows:

1. Matching roughly. The laser data points are projected to the region, and a matched reference model is obtained, and the Gauss blurring is carried out near the grid point of the laser data.

2. Matching precisely, rough matching uses low resolution search to get the mean position of obstacles. The precise matching method is the same as rough matching. The obstacle

location is further divided by higher resolution, and a more precise solution is obtained.

3. Calculating mean and covariance, several position and pose states with the best response are selected to take the average value as the matching result in each matching process. Covariance is divided into position covariance and angle covariance.

5) Loop Closure

Loop closure is to detect whether the current frame matches the previous frame or not. The step is as follow:

1. Find all the adjacent nodes from the map in a certain distance.

2. According to the ID serial number, the node is matched with the node in a certain distance, and the close loop is considered when the mean and covariance of the matching reach a certain requirement.

3. modified the map, add the edge in the closed position of the map to make the map a closed loop.

6) Global Optimization

When a map forms a closed loop, a map needs to be optimized globally used Graph Optimization[14]. The quadrotor's pose in WCS is $c_i = [t_i, \theta] = [x_i, y_i, \theta]^T$. We can calculate any bias of the pose between C_i and C_j by the equation (8).

$$h(c_i, c_j) \equiv \begin{cases} R_i^T (t_i - t_j) \\ \theta_i - \theta_j \end{cases} \quad (8)$$

The R_i^T is a 2×2 coordinate transformation matrix about θ_i . The calculation of the error and the total error equation is shown as follows.

$$e_{ij} \equiv \bar{z}_{ij} - h(c_i, c_j) \quad (9)$$

$$x^2(c, p) = \sum_{ij} e_{ij}^T \Lambda_{ij} e_{ij} \quad (10)$$

The Λ_{ij} is the inverse matrix of covariance. We find the best pose of pose c by minimize the total error x^2 . Levenberg-Marquardt(LM) can be got iteratively.

$$\text{Definite } \Lambda \equiv \begin{bmatrix} \Lambda_{ab} & & \\ & \ddots & \\ & & \Lambda_{mn} \end{bmatrix}, J \equiv \frac{\partial e}{\partial x}, H \equiv J^T \Lambda J$$

The LM system formula is shown as follow.

$$(H + \lambda \text{diag}(H)) \Delta x = J^T \Lambda e \quad (11)$$

Δx can be solved by equation(11), and we bring the Δx to the pose updating equation(12) and (13).

$$t_i = t_i + \Delta t_i \quad (12)$$

$$\theta_i = \theta_i + \Delta \theta_i \quad (13)$$

III. PATH PLANNING WITHIN BELOW-CANOPY ENVIRONMENT

In order to ensure the quadrotor flying safely under the canopy, it must have real time path planning and obstacle avoidance ability. Because of the low computing performance of the airborne computer, the path planning requires higher running speed. To solve the above problems, we propose an

adaptive dynamic step size and extended strategy to increase the compute speed based on the classical RRT*[15] algorithm. The path planning finds an obstacle-free trajectory based on the waypoints generated by the improved RRT* algorithm fitted with the Bezier curve.

A. Description of RRT*

The principle of the RRT* algorithm is to establish the state tree with the initial state as the root node and forward integral[16]. The flow chart of the RRT* algorithm is shown in Table 1.

Table 1 : The flow chart of the RRT*

RRT* Algorithm
1. RRT*Tree(q_{start});
2. while(!ReachTarget)
3. $q_{rand} \leftarrow \text{RANDOM_STATE}()$;
4. $q_{new} \leftarrow \text{STEER}(q_{rand}, q_{near})$;
5. if (ObstacleFree(q_{new})) then
6. $q_{nearest} \leftarrow \text{NEAREST_NEIGHBOR}(q_{new}, r)$;
7. if (ObstacleFree(q_{new})) & COST($q_{nearest}$) < C_{min}
8. Find.Father_node(q_{new});
9. RRT*Tree.AddVertex(q_{new})
10. if (Distance(q_{new}, q_{goal}) < ϵ) then
11. RRT*Path=SearchPath(RRT*Tree);
12. ReachGoal=true;
13. return RRT* Path

B. Improved Strategy of RRT*

There are some disadvantages when using the RRT*:

1).The algorithm uses the global uniform random strategy and the searching path is also random[17].

2).The algorithm uses the fixed step which can not improve the path search efficiency[18] to expand the sampling point.

In order to solve the above problems, we proposed an improved algorithm:

1).The dynamic extended step size is introduced and the feedback information of the environment is dynamically adjusted.

2).An adaptive sampling strategy with biased probability distribution is introduced.

1) Adaptive Extended Step-size

In order to speed up leaving the local minimum area and improve the planning efficiency[19], we introduce the dynamic extended step size strategy according to the different intensity of tree distribution under different canopy areas. Supposing the quadrotor's airborne sensor exploring radius is r , the initial step size is EPS_{init} . When the algorithm generates the current new node q_{new} , we explore the obstacles in the area of a circle with the q_{new} as a center, r as the radius and record the number n_o of the obstacles and their minimum distance d_{min} to q_{new} . If there are several obstacles in the area of circle with a radius of r , we adjust the extended step size EPS according to the equation (14).

$$EPS = \alpha * d_{min} \quad (14)$$

α is an operator less than 1 and changeable. Else if the $n_o = 0$, $EPS = EPS_{init}$

2) Extended Strategy

Every expansion needs to select a point q_{rand} within the space at random. The distribution characteristics of sampling points determine the ability of the algorithm to explore the location area. This paper proposed an extension strategy with target bias. First of all, a probability value p is obtained at random according to the distribution of uniform probability. We set $q_{rand} = q_{goal}$ if $p \geq p_s$. Oppositely, q_{rand} is obtained at the probability of p .

The p_s is threshold that calculated by the equation (15)

$$p_s = n_{near} / n_{all} \quad (15)$$

n_{near} means the number of the obstacles in the area of a circle with the q_{rand} as a center and the r as a radius, n_{all} represents the number of obstacles that have been explored in the area.

3) Path Smoothing Using Bezier Curve

RRT* algorithm generates a series of waypoints, which can not meet the actual flight requirements of the aircraft, and we need smooth the generated path. The waypoints generated by the RRT* are used as the control points of the Bezier curves[20], and a smooth path with continuous curvature can be generated.

$$P(t) = \sum_{i=0}^N P_i B_{i,N}(t) \quad t \in [0,1] \quad (16)$$

Five control points can determine the only four order Bezier curve, and have affine transformation invariance.

q_n is the goal node in the feasible waypoints sequence $\{q_1, \dots, q_n\}$. In this paper, the four order Bezier curve is used to fit the sequence points. The $N+1$ vertex of the Bezier curve can be defined as the N sub polynomial curve, and the parameter equation of each point on the curve is shown as equation (17).

$$P(t) = P_0(1-t)^4 + 4P_1(1-t)^3t + 6P_2(1-t)^2t^2 + 4P_3(1-t)t^3 + P_4t^4 \quad t \in [0,1] \quad (17)$$

The Bezier curves between q_i and q_{i+4} satisfying the equation (18).

$$\begin{aligned} P(0) &= q_i \\ P(1) &= q_{i+4} \end{aligned} \quad (18)$$

In this way, the four order Bezier smooth path can be generated between every 5 points.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. The SLAM Experiment

The simulation experiment of quadrotor SLAM algorithm was carried out through the ROS system and the gazebo robot simulation platform, by using the ROS simulated Hokuyo UTM-30LX laser radar and Logitech C310 camera. Two

cylinders were placed to simulate the environment under the canopy. The center's coordinate of the cylinder is (3,3), (8,8), the radius is 1.5m, and the height is 3M, the result is shown in Fig.3,4 and 5.

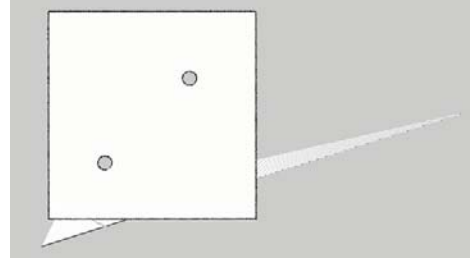


Fig.3 The map of the HectorSLAM simulation

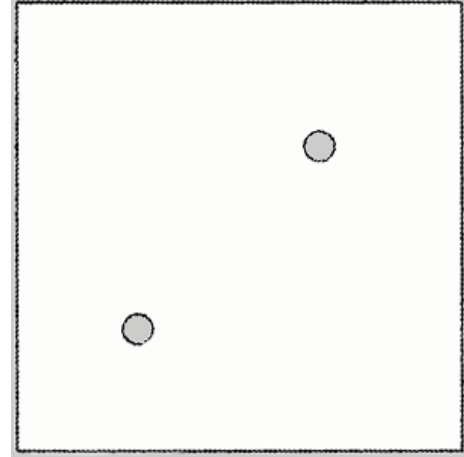


Fig.4 The map of the gmappingSLAM simulation

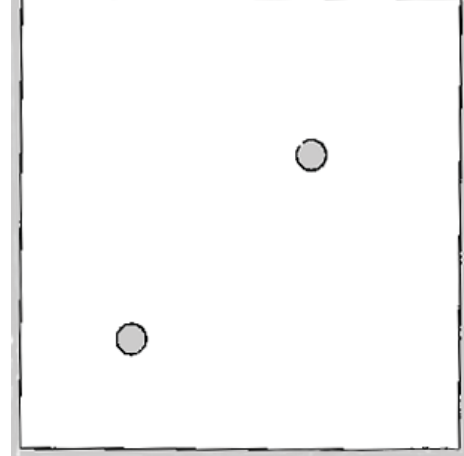


Fig.5 The map of the our algorithm simulation map
The mean and variance of the errors are shown in Table 2 as follows.

Algorithm	Mean	variance
Hector SLAM	0.7234	0.0956
Gmapping SLAM	0.1012	0.0118
Proposed SLAM	0.0901	0.0092

As the Table.2 shows that the proposed SLAM algorithm is more suitable to be used within GPS-denied below-canopy environment due to its less error of mean and variance compare to the Hector_SLAM and Gmapping.

B. The Path Planning Experiment

In order to verify the performance of the proposed path planning algorithm, we simulate the RRT* algorithm and the proposed algorithm 500 times and get their average performance value. Assuming that the quadrotor flies within GPS-denied below-canopy environment with the trunks shaped in cylinder and these obstacles have different position and trunk radius. We took an experiment used the above algorithm. The following figures show the experimental results of path planning using different algorithms. The coordinates of the start point is (20, 30) and the coordinates of the goal is (800,700). The experiment result of classic RRT* and the improved RRT* are shown in the Fig.6 and Fig.7. The deep green point in the figure represents the trunk model in view angle. In the Fig.7, the green line represents the fitted Bezier curve and the red lines represents the path consists of the waypoints generated by the improved algorithm.

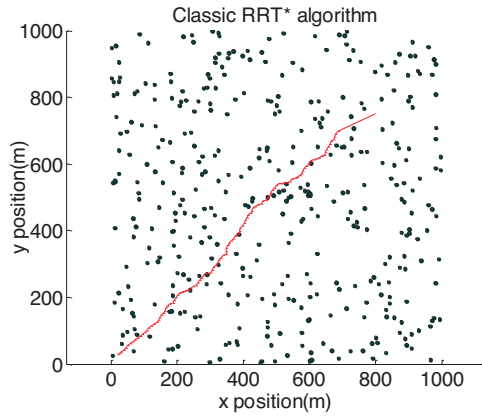


Fig.6 The result of the classic algorithm

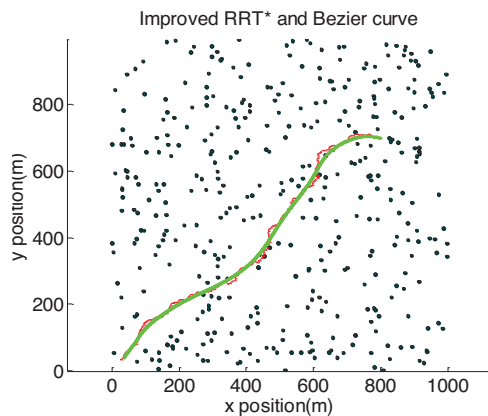


Fig.7 The result of the improved algorithm

We also simulated in the 3D environment with the shape of shaped cylinder shown in Fig8.

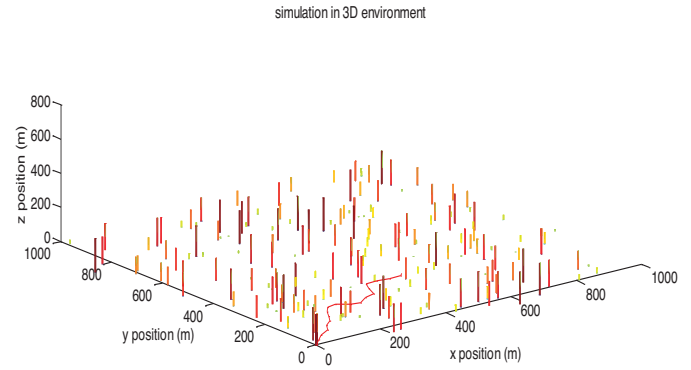


Fig.8 the result of the improved algorithm in 3D environment

The Performance comparison of the planning algorithm is shown in the Table 3

Table 3: Performance comparison of the planning algorithm

500 times simulations	Traditional RRT*	Improved RRT*	Trajectory fitted in Bezier
Numbers of nodes	2309	893	893
Path length	1309	1134	1054
Running time	130s	14s	20s
Failures times	13	5	5

Obviously, the proposed planning algorithm can use less time and generate shorter path than the classic algorithm. The trajectory will decrease the searching failure times so that it will lighten the flight burden for the quadrotor.

V. CONCLUSION

In this paper, the quadrotor navigation system is consisted of the SLAM system and path planning system. This paper proposed a SLAM algorithm based on fusion of monocular vision odometer and laser radar along with an improved RRT* algorithm to generate a series of waypoints used to create a smooth and obstacle-free trajectory fitted with Bezier curves. The experiment results show that the proposed SLAM algorithm is more suitable to be used within GPS-denied below-canopy environment and the proposed path planning algorithm can spend less time to generate a smooth trajectory for the quadrotor to autonomous navigate within GPS-denied below-canopy environment.

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